



2010 Preliminary Course Final Examination

- Working time – $2\frac{1}{2}$ hours.
- Commence each new question on a new page.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

- ☐ 11M3A – Mr Trenwith
- ☐ 11M3B – Mr Fletcher
- ☐ 11M3C – Mr Ireland
- ☐ 11M3D – Mr Lam
- ☐ 11M3E – Mr Rezcallah
- ☐ 11M3F – Mr Weiss
- ☐ 11M3G – Mr Berry/Mr Fletcher

NAME: # PAGES USED:

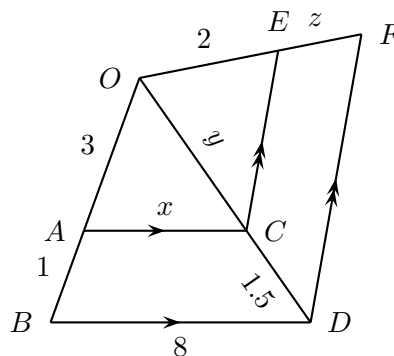
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Question 1 (12 Marks) Commence a NEW page. **Marks**

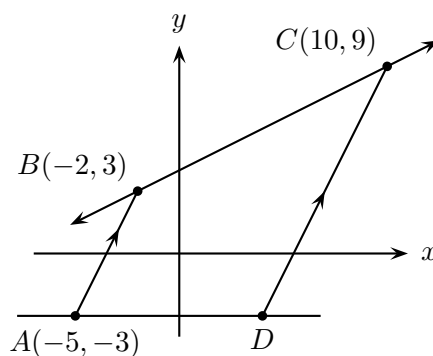
- (a) Evaluate $\frac{\sqrt[3]{2.54 \times 6.78}}{\pi^3}$ correct to 3 significant figures. **2**
- (b) Find integers a and b such that $\frac{\sqrt{3}+1}{2-\sqrt{3}} = a + b\sqrt{3}$. **2**
- (c) Write the exact value of $2 \sin 60^\circ \cos 30^\circ$. **2**
- (d) Solve $y = 2x$ and $y = x^2 - 15$ simultaneously. **3**
- (e) Solve $\frac{1}{x} \leq 4x$. **3**

Question 2 (12 Marks) Commence a NEW page. **Marks**

- (a) Find x , y and z in the diagram, giving all reasons and showing all working. **3**



- (b) On a number plane the points $A(-5, -3)$, $B(-2, 3)$, $C(10, 9)$ and D form a trapezium, in which AB is parallel to DC . AD is parallel to the x axis.



- Show that the equation of DC is $2x - y - 11 = 0$. **2**
- Find the coordinates of the point D . **2**
- Find the angle that the line DC makes with the positive x axis. **1**
- Find the equation of the circle centred at C with radius BC . **2**
- Find the coordinates of the point M that divides BC externally in the ratio $2 : 3$. **2**

Question 3 (12 Marks)	Commence a NEW page.	Marks
(a) Differentiate with respect to x :		
i. $f(x) = x\sqrt{x}$.		1
ii. $g(x) = (3x^2 + 4)^3$.		2
iii. $h(x) = \frac{x^2}{x^2 + 9}$.		2
iv. $y = x^5(3 - x^2)^8$. Fully factorise your answer to this part.		3
(b) Evaluate $\lim_{x \rightarrow -3} \frac{x^3 + 27}{x + 3}$.		2
(c) Sketch the curve with the following properties:		2
• $f'(x) < 0$ and $f''(x) < 0$ when $-2 < x < 0$. • $f(0) = 0$. • $f'(x) > 0$ and $f''(x) < 0$ when $0 < x < 2$.		

Question 4 (12 Marks)	Commence a NEW page.	Marks
(a) At what points on the curve $y = x^3 - 4x^2 + 2x$ are tangents parallel to the line $2x + y = 3$?		3
(b) For the curve $y = x^5 - x^4$		
i. Find the x intercepts of the curve.		1
ii. Find and classify all stationary points of the curve.		3
iii. Find the x coordinates of all points of inflexion.		3
iv. Hence, sketch the graph, showing all important features.		2

Question 5 (12 Marks)

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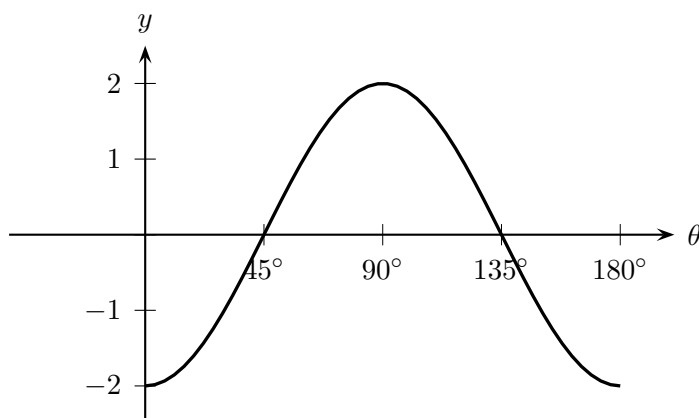
Marks

(a) Solve $2 \sin^2 x - \sin x = 0$ for $0^\circ \leq x \leq 360^\circ$. **3**

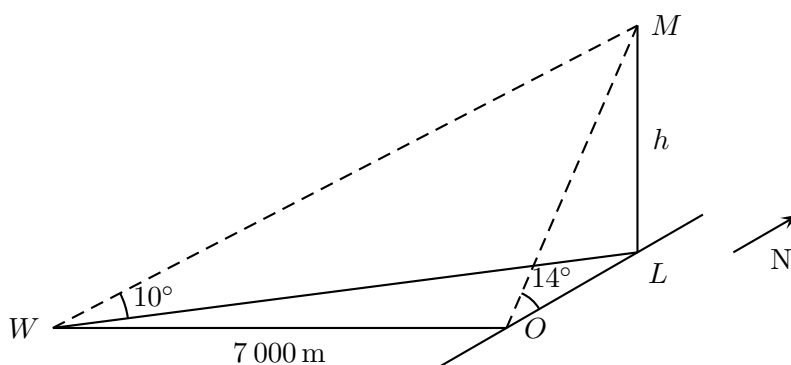
- (b) A ship S sails from a point P on a bearing of $N60^\circ E$ for 56 nm. Ship B leaves port P on a bearing of $110^\circ T$ for 48 nm. **3**

Calculate the direct distance from S to B , correct to the nearest nautical mile.

- (c) The diagram below shows part of a sine or cosine curve between 0° and 180° .



- i. State the amplitude a . **1**
 - ii. State the period T . **1**
 - iii. Write down the equation of the curve. **1**
- (d) The angle of elevation of the summit of a mountain due north of O is 14° . Upon walking 7 000 m due west, it is found to be 10° .



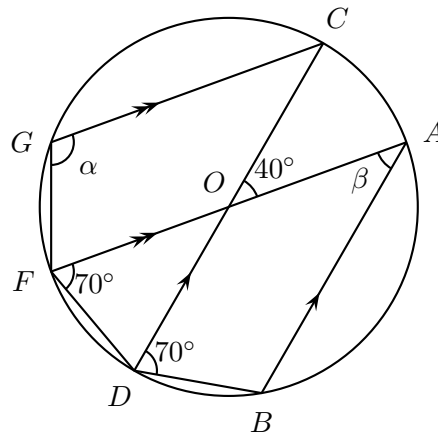
- i. Express WL in terms of h . **1**
- ii. Find the height of the mountain correct to the nearest metre. **2**

Question 6 (12 Marks)

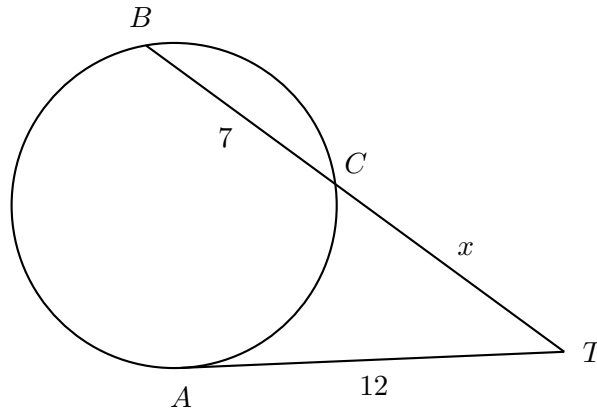
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Marks

- (a) Find the value of α and β , giving full reasons.

4

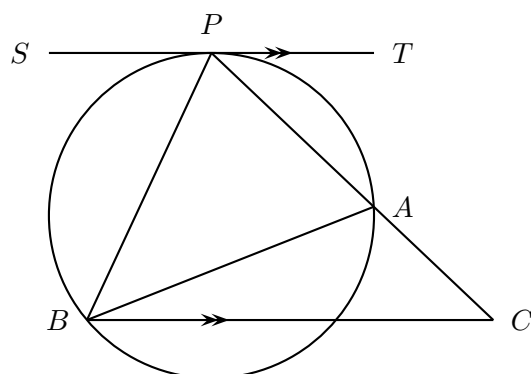
- (b) The line AT is the tangent to the circle at A , and BT is a secant meeting the circle at B and C .

2

Given that $AT = 12$, $BC = 7$ and $CT = x$, find the value of x (giving all reasons).

Question 6 continues on the next page ...

- (c) In the diagram, A , P and B are points on the circle. The line PT is tangent to the circle at P , and PA is produced to C so that BC is parallel to PT .

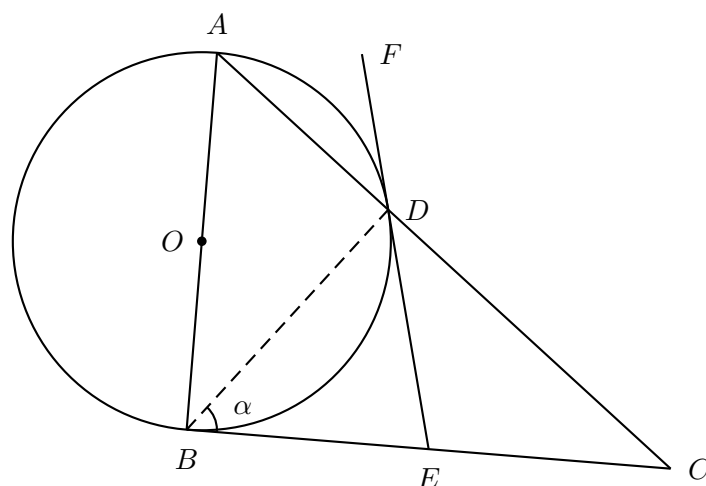


Copy the diagram on to your paper.

- i. Show that $\angle PBA = \angle PCB$. **2**
- ii. By showing $\triangle PBA \parallel \triangle PBC$, deduce that $PB^2 = PA \times PC$. **2**

- (d) In the diagram, AB is the diameter of the circle with centre O . BC is a tangent to the circle at B . The line AC is a straight line and intersects the circle at D . The tangent to the circle at D intersects BC at E . Let $\angle EBD = \alpha$. **2**

Reproduce the diagram on to your page.



Prove $\angle EDC = 90^\circ - \alpha$.

Question 7 (12 Marks)

Commence a NEW page.

Marks

- (a) i. Show the equation **1**

$$x - 2y - 4 + k(3x - y + 1) = 0$$

(where k is a constant)

can be rewritten as

$$(1 + 3k)x - (2 + k)y + (k - 4) = 0$$

- ii. Hence or otherwise, find the equation of a straight line passing through the intersection of $x - 2y - 4 = 0$, $3x - y + 1 = 0$ and perpendicular to the line $4x + 5y + 6 = 0$. **3**

- (b) By using the perpendicular distance formula or otherwise, find all possible values of a if $3x + 4y + a = 0$ is a tangent to the circle **3**

$$x^2 + (y + 1)^2 = 9$$

- (c) Sketch the region defined by $y \geq \frac{1}{x}$. (Hint: be very careful!) **2**

- (d) i. Sketch the graph of $y = |x - 2|$. **1**

- ii. For what values of x is $|x - 2| < \frac{1}{2}x$? **2**

Question 8 (12 Marks)

Commence a NEW page.

Marks

- (a) i. For what values of x does this series have a limiting sum? **2**

$$2x + 3 + (2x + 3)^2 + (2x + 3)^3 + \dots$$

- ii. Find that limiting sum in terms of x . **1**

- (b) Solve the equation $2 \log_3 x = \log_3(6 - x)$. Show all necessary working. **3**

- (c) i. Rewrite $2^x = 5^y$ with $\frac{x}{y}$ as the subject by taking logarithms to a suitable base. **1**

- ii. Hence or otherwise, find the exact value of $8^{\frac{x}{y}}$ if $2^x = 5^y$. **2**

- (d) Boxes in a storeroom are stacked in a pile such that there are 25 on the bottom, 22 on the next, 19 afterwards etc, until 116 boxes are on the pile together.

- i. How many rows of boxes are there? **2**

- ii. How many boxes are there on the top row? **1**

Question 9 (12 Marks)

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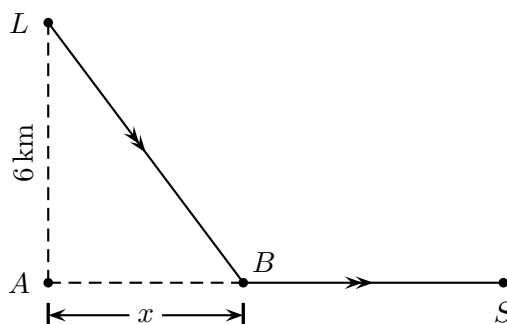
Marks

- (a) A particle moves in a straight line such that its displacement in meters, x , after t seconds is given by

$$x = t^3 + 3t^2 - 9t - 1$$

- | | | |
|------|--|----------|
| i. | Find the velocity of the particle at any time t . | 1 |
| ii. | When and where does the particle first change direction? | 2 |
| iii. | What is the average speed of the particle in the first second? | 1 |

- (b) The water's edge is in a straight line ABS which runs east-west. A lighthouse is 6 km due north of A .



10 km due east of A is the general store (S). To travel from the lighthouse (L) to the general store as quickly as possible, the lighthouse keeper rows from L to B , which is x km from A and then jogs to the general store. The lighthouse keeper's rowing speed is 6 km/h and his jogging speed is 10 km/h.

- | | | |
|-----|--|----------|
| i. | Show that it takes the lighthouse keeper $\frac{\sqrt{36+x^2}}{6}$ hours to row from the lighthouse to B . | 2 |
| ii. | Show that the total time taken for the lighthouse keeper to reach the general store is given by | 2 |

$$t = \frac{\sqrt{36+x^2}}{6} + \frac{10-x}{10} \text{ hours}$$

- | | | |
|------|---|----------|
| iii. | Hence by using calculus, show that when $x = 4.5$ km, the time that it takes for the lighthouse keeper to travel from the lighthouse to the general store is a minimum. | 3 |
| iv. | Hence find the quickest time it takes to the lighthouse keeper to go to the general store from the lighthouse. Give your answer correct to the nearest minute. | 1 |

End of paper.

Suggested Solutions

Question 1 (Lam – starts page 2)

(a) (2 marks)

- ✓ [1] for correct numerical calculation.
- ✓ [1] for correct rounding.

$$\frac{\sqrt[3]{2.54 \times 6.78}}{\pi^3} = 0.083 \text{ 3 (3 s.f.)}$$

(b) (2 marks)

- ✓ [1] for multiplying by $\frac{x+2\sqrt{3}}{x+2\sqrt{3}}$.
- ✓ [1] for correct final answer.

$$\begin{aligned} \frac{\sqrt{3}+1}{2-\sqrt{3}} \times \frac{x+2\sqrt{3}}{x+2\sqrt{3}} &= \frac{(\sqrt{3}+1)(2+\sqrt{3})}{4-3} \\ &= 2\sqrt{2} + (\sqrt{3})^2 + 2 + \sqrt{3} \\ &= 5 + 3\sqrt{3} \\ \therefore a &= 5 \quad b = 3 \end{aligned}$$

(c) (2 marks)

- ✓ [1] for correct exact ratios of $\sin 60^\circ$ and $\cos 30^\circ$.
- ✓ [1] for final answer.

$$\begin{aligned} 2 \sin 60^\circ \cos 30^\circ &= 2 \times \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} \\ &= \frac{3}{2} \end{aligned}$$

(d) (3 marks)

- ✓ [1] for equating the given equations.
- ✓ [1] for x values.
- ✓ [1] for y values.

$$\begin{aligned} \begin{cases} y = 2x \\ y = x^2 - 15 \end{cases} \\ x^2 - 15 &= 2x \\ x^2 - 2x - 15 &= 0 \\ (x-5)(x+3) &= 0 \\ \therefore x &= 5, -3 \\ \therefore y &= 10, -6 \end{aligned}$$

(e) (3 marks)

- ✓ [1] for multiplying throughout by x^2 .
- ✓ [1] for correct $-\frac{1}{2} \leq x < 0$.
- ✓ [1] for correct $x \geq \frac{1}{2}$.

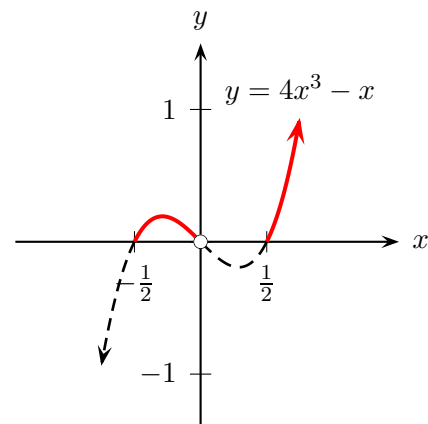
$$\frac{1}{x} \leq \frac{4x}{x^2}$$

$$x \leq 4x^3$$

$$4x^3 - x \geq 0$$

$$x(4x^2 - 1) \geq 0$$

$$x(2x-1)(2x+1) \geq 0$$

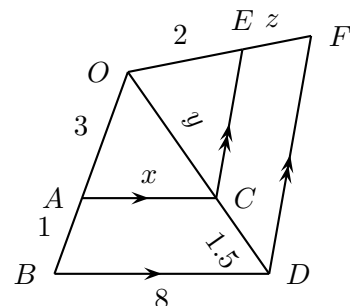


$$\therefore -\frac{1}{2} \leq x < 0 \text{ or } x \geq \frac{1}{2}$$

Question 2 (Lam – starts page 2)

(a) (3 marks)

- ✓ [1] for each correct pronumeral found with reasons. If insufficient reasoning is given, a maximum of one mark will be lost.



- By similar $\triangle OAC$ and $\triangle OBD$,

$$\begin{aligned}\frac{OA}{OB} &= \frac{AC}{BD} \\ \frac{3}{4} &= \frac{x}{8} \\ \therefore x &= 6\end{aligned}$$

- By the intercepts of transversals,

$$\begin{aligned}\frac{OA}{OB} &= \frac{OC}{OD} = \frac{OE}{EF} \\ \therefore \frac{3}{1} &= \frac{y}{1.5} = \frac{2}{z}\end{aligned}$$

Take the first pair of fractions,

$$y = 1.5 \times 3 = 4.5$$

Take the first and third pair of fractions,

$$\begin{aligned}\frac{2}{z} &= 3 \\ z &= \frac{2}{3}\end{aligned}$$

$$\therefore x = 6, y = 4.5, z = \frac{2}{3}$$

- (b) i. (2 marks)

✓ [1] for realising $m_{DC} = m_{AB}$ and finding m_{AB} .

✓ [1] for showing $y = 2x - 11$ or $2x - y - 11 = 0$.

As $AB \parallel DC$, then $m_{AB} = m_{DC}$.

$$\begin{aligned}m_{DC} &= m_{AB} \\ &= \frac{3 - (-3)}{-2 - (-5)} = \frac{6}{3} = 2\end{aligned}$$

Apply the point-gradient formula with $C(10, 9)$:

$$\begin{aligned}y - y_1 &= m(x - x_1) \\ y - 9 &= 2(x - 10) \\ y &= 2x - 20 + 9 \\ y &= 2x - 11 \\ 2x - y - 11 &= 0\end{aligned}$$

- ii. (2 marks)

✓ [1] for using y coordinate is -3 .

✓ [1] for $D(4, -3)$.

D has the same y coordinate as $A(-5, -3)$. Hence

$$2x - (-3) - 11 = 0$$

$$2x + 3 - 11 = 0$$

$$2x = 8$$

$$x = 4$$

$$\therefore D(4, -3)$$

- iii. (1 mark)

✓ [Note:] Accept 63° , 63.43° , $63^\circ 26'$.

$$m_{DC} = 2 = \tan \theta$$

$$\theta = \tan^{-1} 2 = 63.43^\circ$$

- iv. (2 marks)

✓ [1] for correct distance of BC .

✓ [1] for final answer.

$$\begin{aligned}d_{BC} &= \sqrt{(10 - (-2))^2 + (9 - 3)^2} \\ &= \sqrt{12^2 + 6^2} = \sqrt{180}\end{aligned}$$

$\therefore$ equation of circle is

$$(x - 10)^2 + (y - 9)^2 = 180$$

- v. (2 marks)

✓ [1] for each correct x and y value.

$$\begin{array}{ccc} & B(-2, 3) & C(10, 9) \\ & \swarrow \quad \searrow & \\ -2 & & 3 \end{array}$$

$$\begin{aligned}M &\left(\frac{\ell x_1 + k x_2}{k + \ell}, \frac{\ell y_1 + k y_2}{k + \ell} \right) \\ &= \left(\frac{3(-2) + (-2)(10)}{-2 + 3}, \frac{3(3) + (-2)(9)}{-2 + 3} \right) \\ &= (-6 - 20, 9 - 18) = (-26, -9)\end{aligned}$$

Question 3 (Ireland – starts on page 3)

(a) i. (1 mark)

$$f(x) = x\sqrt{x} = x^{\frac{3}{2}}$$

$$f'(x) = \frac{3}{2}x^{\frac{1}{2}}$$

ii. (2 marks)

✓ [1] for applying chain rule.

✓ [1] for final answer.

$$g(x) = (3x^2 + 4)^3$$

$$\left| \begin{array}{ll} g(u) = u^3 & u(x) = 3x^2 + 4 \\ g'(u) = 3u^2 & u'(x) = 6x \end{array} \right.$$

$$g'(x) = g'(u) \times u'(x)$$

$$= 3u^2 \times 6x$$

$$= 3(3x^2 + 4)^2 \times 6x$$

$$= 18x(3x^2 + 4)^2$$

iii. (2 marks)

✓ [1] for applying quotient rule.

✓ [1] for final answer.

$$h(x) = \frac{x^2}{x^2 + 9}$$

$$\left| \begin{array}{ll} u = x^2 & v = x^2 + 9 \\ u' = 2x & v' = 2x \end{array} \right.$$

$$h'(x) = \frac{vu' - uv'}{v^2}$$

$$= \frac{2x(x^2 + 9) - 2x(x^2)}{(x^2 + 9)^2}$$

$$= \frac{18x}{(x^2 + 9)^2}$$

iv. (3 marks)

✓ [1] for applying product rule.

✓ [1] for applying chain rule.

✓ [1] for final fully factorised answer.

$$y = x^5(3 - x^2)^8$$

$$\left| \begin{array}{ll} u = x^5 & v = (3 - x^2)^8 \\ u' = 5x^4 & v' = 8 \times (-2x) \times (3 - x^2)^7 \\ & = -16x(3 - x^2)^7 \end{array} \right.$$

$$\frac{dy}{dx} = uv' + vu'$$

$$= -16x^6(3 - x^2)^7 + 5x^4(3 - x^2)^8$$

$$= x^4(3 - x^2)^7(-16x^2 + 5(3 - x^2))$$

$$= x^4(3 - x^2)^7(-21x^2 + 15)$$

$$= 3x^4(3 - x^2)^7(-7x^2 + 5)$$

(b) (2 marks)

✓ [1] for correct factorisation of numerator.

✓ [1] for final answer.

$$\lim_{x \rightarrow -3} \frac{x^3 + 27}{x + 3}$$

$$= \lim_{x \rightarrow -3} \frac{(x+3)(x^2 - 3x + 9)}{x+3}$$

$$= \lim_{x \rightarrow -3} x^2 - 3x + 9$$

$$= (-3)^2 - 3(-3) + 9$$

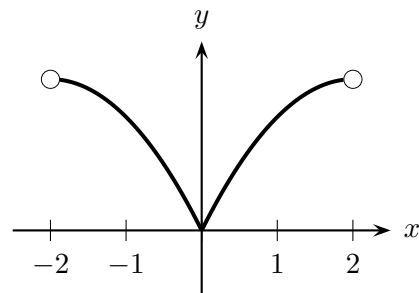
$$= 27$$

(c) (2 marks)

✓ [1] for shape.

✓ [1] for $f(0) = 0$.• $f'(x) < 0$ & $f''(x) < 0$ ($-2 < x < 0$).• $f(0) = 0$.• $f'(x) > 0$ & $f''(x) < 0$ ($0 < x < 2$).

One possibility:



Question 4 (Ireland – starts on page 3)

(a) (3 marks)

- ✓ [1] for correctly differentiating.
- ✓ [1] solve for x ; obtaining $x = 2$ & $x = \frac{2}{3}$.
- ✓ [1] for finding corresponding y values.

$$y = x^3 - 4x^2 + 2x$$

$$\frac{dy}{dx} = 3x^2 - 8x + 2$$

The gradient of the line $2x + y = 3$ is -2 .
The tangent to the curve is parallel to the line when $\frac{dy}{dx} = -2$:

$$3x^2 - 8x + 2 = -2$$

$$3x^2 - 8x + 4 = 0$$

$$(3x - 2)(x - 2) = 0$$

$$\therefore x = 2, \frac{2}{3}$$

When $x = 2$,

$$y = 2^3 - 4(2^2) + 2(2) \\ = 8 - 16 + 4 = -4$$

When $x = \frac{2}{3}$,

$$y = \left(\frac{2}{3}\right)^3 - 4\left(\frac{2}{3}\right)^2 + 2\left(\frac{2}{3}\right) = -\frac{4}{27}$$

Hence the points on the curve parallel to $2x + y = 3$ are

$$(2, -4) \quad \& \quad \left(\frac{2}{3}, -\frac{4}{27}\right)$$

(b) i. (1 mark)

$$x^5 - x^4 = 0$$

$$x^4(x - 1) = 0$$

$$\therefore x = 0, 1$$

ii. (3 marks)

- ✓ [1] for correct differentiation.
- ✓ [1] for concluding $(0, 0)$ is a local max.
- ✓ [1] for concluding $\left(\frac{4}{5}, -\frac{256}{3125}\right)$ is a local minimum.

- ✓ [Note:] maximum 1 mark will be lost for not finding the y coordinates.

$$\frac{dy}{dx} = 5x^4 - 4x^3$$

Stationary pts occur when $\frac{dy}{dx} = 0$:

$$5x^4 - 4x^3 = 0$$

$$x^3(5x - 4) = 0$$

$$\therefore x = 0, \frac{4}{5}$$

- At $x = 0$, $y = 0$.
- At $x = \frac{4}{5}$,

$$y = \left(\frac{4}{5}\right)^5 - \left(\frac{4}{5}\right)^4 = -\frac{256}{3125}$$

x	-1	0	$\frac{1}{2}$	$\frac{4}{5}$	1
y'	+	0	-	0	+
y					

Hence $x = 0$ is a local maximum and $x = \frac{4}{5}$ is a local minimum.

iii. (3 marks)

- ✓ [1] for correct differentiation.
- ✓ [1] for $x = \frac{3}{5}$ being a point of inflexion.
- ✓ [1] testing possible inflexions to eliminate $x = 0$ and confirm $x = \frac{3}{5}$.

$$\frac{d^2y}{dx^2} = 20x^3 - 12x^2$$

Pts of inflexion occur when $\frac{d^2y}{dx^2} = 0$:

$$20x^3 - 12x^2 = 0$$

$$4x^2(5x - 3) = 0$$

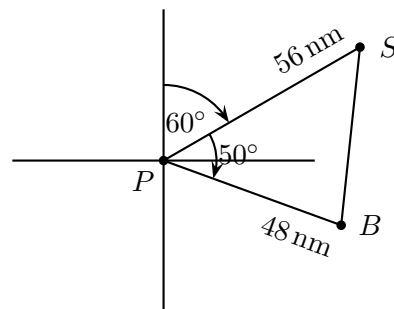
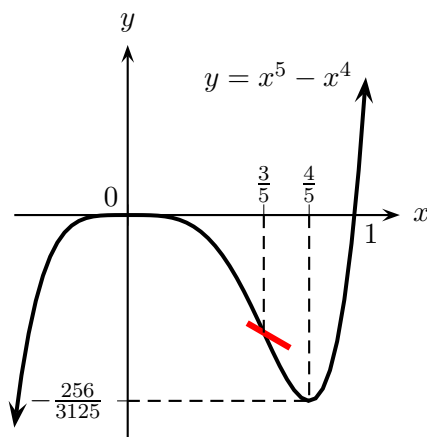
$$\therefore x = 0, \frac{3}{5}$$

x	-1	0	$\frac{1}{2}$	$\frac{3}{5}$	1
y''	-	0	-	0	+
y'					

As the sign of the second derivative does not change around $x = 0$, it is therefore not a point of inflexion despite $f''(0) = 0$. Hence the only point of inflexion occurs at $x = \frac{3}{5}$.

iv. (2 marks)

- ✓ [2] for correct shape with details, provided it follows on from previous working.
- ✓ [-1] for each error. If the arrowheads of the end of the curve point in the same direction, deduct [1].



$$\angle SPB = 110^\circ - 60^\circ = 50^\circ$$

Apply the cosine rule in $\triangle PSB$,

$$\begin{aligned} SB^2 &= 56^2 + 48^2 - 2(56)(48) \cos 50^\circ \\ &= 1\,984.37 \end{aligned}$$

$$\therefore SB = 44.55 \dots = 45 \text{ nm (nearest nm)}$$

- (c) i. (1 mark) $a = 2$
- ii. (1 mark) $T = 180^\circ$
- iii. (1 mark) $y = -2 \cos 2\theta$
- (d) i. (1 mark)

Question 5 (Fletcher – starts on page 4)

(a) (3 marks)

- ✓ [1] for factorising to $\sin x(2 \sin x - 1) = 0$.
- ✓ [1] each for the solutions: $x = 0^\circ, 180^\circ, 360^\circ$.
- ✓ [1] each for the solutions: $x = 30^\circ, 150^\circ$.

$$\begin{aligned} 2 \sin^2 x - \sin x &= 0 \\ \sin x(2 \sin x - 1) &= 0 \\ \sin x = 0 &\quad \left| \quad \sin x = \frac{1}{2} \right. \\ x = 0^\circ, 180^\circ, 360^\circ &\quad \left| \quad x = 30^\circ, 150^\circ \right. \\ \therefore x = 0^\circ, 30^\circ, 150^\circ, 180^\circ, 360^\circ \end{aligned}$$

(b) (3 marks)

- ✓ [1] for correct $\triangle PSB$ with $\angle SPB = 50^\circ$.
- ✓ [1] for applying cosine rule.
- ✓ [1] for final answer. Do not penalise for rounding.

$$\begin{aligned} \frac{h}{WL} &= \tan 10^\circ \\ \therefore WL &= \frac{h}{\tan 10^\circ} \end{aligned} \quad (5.1)$$

ii. (2 marks)

- ✓ [1] for obtaining $OL^2 + 7\,000^2 = WL^2$.
- ✓ [1] for final answer.

$$\begin{aligned} \frac{h}{OL} &= \tan 14^\circ \\ \therefore OL &= \frac{h}{\tan 14^\circ} \end{aligned} \quad (5.2)$$

In $\triangle WOL$, $\angle WOL = 90^\circ$ as W is due west of O . Hence

$$OL^2 + 7\,000^2 = WL^2 \quad (5.3)$$

Substitute (5.1) and (5.2) into (5.3) (c) i. (2 marks)

$$\begin{aligned}
 WL^2 - OL^2 &= 7\,000^2 \\
 \left(\frac{h}{\tan 10^\circ}\right)^2 - \left(\frac{h}{\tan 14^\circ}\right)^2 &= 7\,000^2 \\
 h^2 \left(\frac{1}{\tan^2 10^\circ} - \frac{1}{\tan^2 14^\circ}\right) &= 7\,000^2 \\
 h^2 \left(\frac{\tan^2 14^\circ - \tan^2 10^\circ}{\tan^2 10^\circ \tan^2 14^\circ}\right) &= 7\,000^2 \\
 h^2 &= \frac{7\,000^2 \tan^2 10^\circ \tan^2 14^\circ}{\tan^2 14^\circ - \tan^2 10^\circ} \\
 h &= 1\,745.8 \text{ m} = 1\,746 \text{ m (nearest m)}
 \end{aligned}$$

Question 6 (Berry – starts on page 5)

(a) (4 marks)

- ✓ [1] for each correct pronumeral found.
- ✓ [1] for working & leading to correct answer.
- To find α :
 - $\angle FOD = 40^\circ$ (vert. opp)
 - $\angle DOA = 180^\circ - 40^\circ = 140^\circ$ (supplementary)
 - $\angle FOC$ (reflex) $= 140^\circ + 80^\circ = 220^\circ$.
 - $\therefore \alpha = 110^\circ$
(angle at the circumference is half the angle at the centre subtended by the same arc)
- To find β :
 - $\beta = 40^\circ$ (alternate $\angle$, $OC \parallel AB$).
 - Alternatively, use $\angle ODF = 70^\circ$ (base $\angle$ of isos $\triangle$ and $\beta = 180^\circ - 140^\circ - 40^\circ$ (opp $\angle$ of cyclic quad).

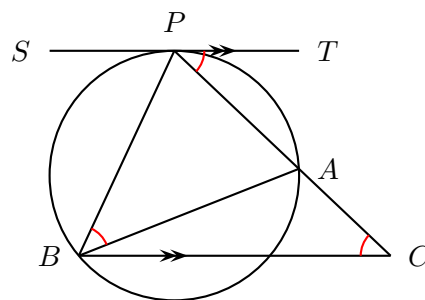
(b) (2 marks)

- ✓ [1] for correctly applying tangent-secant theorem.
- ✓ [1] for solving and obtaining $x = 9$.
- $x(x + 7) = 12^2$ (tangent-secant thm)

$$\begin{aligned}
 x^2 + 7x - 144 &= 0 \\
 (x - 9)(x + 16) &= 0 \\
 x &= 9, -16
 \end{aligned}$$

- $x = -16$ is not a valid solution as x is a length. Hence $x = 9$.

✓ [1] for each correct reason.



- $\angle TPC = \angle PCB$
(alternate $\angle$, $PT \parallel BC$)
- $\angle TPC = \angle PBA$
($\angle$ in alternate segment)

$$\therefore \angle PBA = \angle PCB.$$

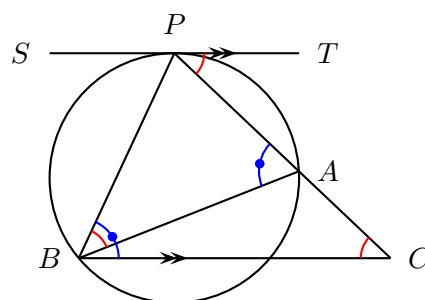
ii. (2 marks)

- ✓ [1] for showing $\triangle PBA \parallel \triangle PCB$ correctly.
- ✓ [1] for $\frac{PB}{PC} = \frac{PA}{PB}$, leading correct solution.

In $\triangle PBA$ & $\triangle PBC$

- $\angle PBA = \angle PBC$
(previously proven)
- $\angle BPA$ is common.
- By the $\angle$ sum of $\triangle$,
 $\angle PAB = \angle PBC$ (remaining $\angle$)

$$\therefore \triangle PBA \parallel \triangle PBC \text{ (equiangular).}$$

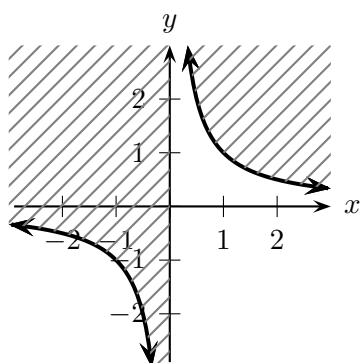


Hence the ratio of corresponding sides are equal, i.e.

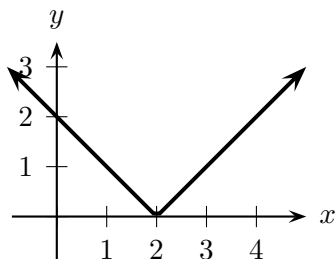
$$\begin{aligned}
 \frac{PB}{PC} &= \frac{PA}{PB} \\
 \therefore PB^2 &= PA \times PC
 \end{aligned}$$

(c) (2 marks)

✓ [1] for each correct region.



(d) i. (1 mark)



ii. (2 marks)

✓ [1] for correct method leading to correct answer.

✓ [1] for obtaining correct interval.

$$\begin{array}{l|l}
 |x - 2| < \frac{1}{2}x & \\
 \hline
 x - 2 < \frac{1}{2}x & -(x - 2) < \frac{1}{2}x \\
 \frac{1}{2}x < 2 & -x + 2 < \frac{1}{2}x \\
 x < 4 & \frac{3}{2}x > 2 \\
 & x > \frac{4}{3} \\
 \hline
 \therefore \frac{4}{3} < x < 4
 \end{array}$$

Alternative method exists: find the points of intersection and use a sketch of $y = \frac{1}{2}x$ and $y = |x - 2|$ to determine the correct interval.

Question 8 (Trenwith – starts on page 7)

(a) i. (2 marks)

✓ [1] for $x > -2$ & $x < -1$.✓ [1] for combining the inequalities ($-2 < x < -1$)

$$r = 2x + 3$$

A limiting sum exists when $|r| < 1$, i.e.

$$-1 < 2x + 3 < 1$$

$$-4 < 2x < -2$$

$$-2 < x < -1$$

ii. (1 mark)

$$\begin{aligned}
 S &= \frac{a}{1 - r} \\
 &= \frac{2x + 3}{1 - (2x + 3)} \\
 &= \frac{2x + 3}{-2x - 2}
 \end{aligned}$$

(b) (3 marks)

✓ [2] for $x = 2, -3$.✓ [1] for discarding $x = -3$ as it is invalid.

$$2 \log_3 x = \log_3 (6 - x)$$

$$\log_3 x^2 = \log_3 (6 - x)$$

$$x^2 = 6 - x$$

$$x^2 + x - 6 = 0$$

$$(x - 2)(x + 3) = 0$$

$$\therefore x = 2, -3$$

When $x = -3$, $2 \log_3 -3$ is not defined.

$$\therefore x = 2$$

(c) i. (1 mark)

$$2^x = 5^y$$

$$\frac{x}{\div y} \log_2 2 = \frac{y}{\div y} \log_2 5$$

$$\therefore \frac{x}{y} = \log_2 5$$

ii. (2 marks)

- ✓ [1] for using $\frac{x}{y}$ as the exponent base 2.
 ✓ [1] for final answer.

$$\begin{aligned}\frac{x}{y} &= \log_2 5 \\ 2^{x/y} &= 2^{\log_2 5} = 5 \\ (2^3)^{\frac{x}{y}} &= \left(2^{\frac{x}{y}}\right)^3 = 5^3 = 125\end{aligned}$$

(d) i. (2 marks)

- ✓ [1] for $3n^2 - 53n + 232 = 0$.
 ✓ [1] for $n = 8$ and justifying $n = \frac{29}{3}$ is invalid. NB. This mark not awarded if $n = 8$ is obtained by guessing/with insufficient working/writing out terms of AP to find 8th term.

$$\begin{aligned}25 + 22 + 19 + \dots \\ a = 25 \quad d = -3 \quad n = ? \quad S_n = 116\end{aligned}$$

Sum of AP formula:

$$\begin{aligned}S_n &= \frac{n}{2}(2a + d(n-1)) \\ 116 &= \frac{n}{2}(2 \times 25 - 3(n-1)) \\ 232 &= n(50 - 3n + 3) \\ 232 &= n(53 - 3n) \\ 3n^2 - 53n + 232 &= 0\end{aligned}$$

Apply the quadratic formula,

$$\begin{aligned}n &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{53 \pm \sqrt{53^2 - 4(3)(232)}}{2 \times 3} = \frac{53 \pm \sqrt{25}}{6} \\ &= 8, \frac{29}{3}\end{aligned}$$

As the number of rows an integer, $n = \frac{29}{3}$ is not possible. (This question is not asking what number of rows will exceed a certain sum, but rather asking for the exact number of rows). Hence $n = 8$.

ii. (1 mark)

$$\begin{aligned}T_n &= a + d(n-1) \\ T_8 &= 25 - 3(8-1) \\ &= 25 - 21 = 4\end{aligned}$$

There are 4 boxes on the top row.

Question 9 (Weiss – starts on page 8)

(a) i. (1 mark)

$$\begin{aligned}x(t) &= t^3 + 3t^2 - 9t - 1 \\ x'(t) &= 3t^2 + 6t - 9\end{aligned}$$

ii. (2 marks)

- ✓ [1] for $t = -3, 1$ and discarding $t = -3$.
 ✓ [1] for finding $x = -6$ m.
 The particle changes direction when $x'(t) = 0$.

$$\begin{aligned}3t^2 + 6t - 9 &= 0 \\ 3(t^2 + 2t - 3) &= 0 \\ (t+3)(t-1) &= 0 \\ \therefore t &= 1, -3\end{aligned}$$

As $t \geq 0$, then $t = -3$ is not valid.

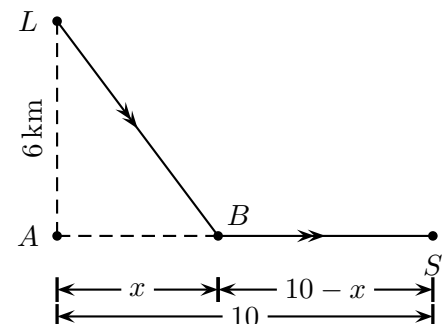
$$x(1) = 1 + 3 - 9 - 1 = -6 \text{ m}$$

iii. (1 mark)

$$\begin{aligned}x(1) &= -6 \quad x(0) = -1 \\ \text{average speed} &= \left| \frac{x(t_2) - x(t_1)}{t_2 - t_1} \right| \\ &= \left| \frac{(-6) - (-1)}{1 - 0} \right| \\ &= 5 \text{ ms}^{-1}\end{aligned}$$

i. (2 marks)

- ✓ [1] for $LB = \sqrt{x^2 + 6^2}$.
 ✓ [1] for $t = \frac{\sqrt{x^2 + 6^2}}{6}$.



In $\triangle LAB$,

$$AL^2 + AB^2 = LB^2$$

$$LB = \sqrt{x^2 + 6^2}$$

As the keeper's rowing speed is 6 km/h,

$$s = \frac{d}{t}$$

$$6 = \frac{\sqrt{x^2 + 6^2}}{t} \Rightarrow t = \frac{\sqrt{x^2 + 6^2}}{6}$$

ii. (2 marks)

✓ [1] for time taken from B to S being $\frac{10-x}{10}$.

✓ [1] for $t = \frac{\sqrt{x^2 + 6^2}}{6} + \frac{10-x}{10}$.

Between B and S ,

$$s = \frac{d}{t}$$

$$10 = \frac{10-x}{t} \Rightarrow t = \frac{10-x}{10}$$

$$\therefore t_{\text{total}} = \frac{\sqrt{x^2 + 6^2}}{6} + \frac{10-x}{10}$$

iii. (3 marks)

✓ [1] for correctly differentiating.

✓ [1] for $x = \pm \frac{9}{2}$.

✓ [1] for discarding invalid solution and verify $x = \frac{9}{2}$ gives a minimum. This mark is lost if the solution is not verified.

Need to minimise t w.r.t. x :

$$t(x) = \frac{1}{6} (x^2 + 6^2)^{\frac{1}{2}} + \frac{1}{10} (10 - x)$$

$$\begin{aligned} t'(x) &= \frac{1}{2} \times \frac{1}{6} \times 2x (x^2 + 6^2)^{-\frac{1}{2}} - \frac{1}{10} \\ &= \frac{x}{6\sqrt{x^2 + 6^2}} - \frac{1}{10} \end{aligned}$$

Minimum time occurs when $t'(x) = 0$, i.e.

$$\frac{x}{6\sqrt{x^2 + 6^2}} - \frac{1}{10} = 0$$

$$\frac{x}{6\sqrt{x^2 + 6^2}} = \frac{1}{10}$$

$$5x = 3\sqrt{x^2 + 6^2}$$

Square both sides,

$$25x^2 = 9x^2 + 324$$

$$\frac{16x^2}{\div 16} = \frac{324}{\div 16}$$

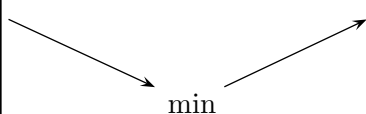
$$x^2 = \frac{81}{4}$$

$$\therefore x = \pm \frac{9}{2}$$

As x is a distance, then $x = -\frac{9}{2}$ is not valid. Check that $x = \frac{9}{2}$ is a local minimum:

$$t'(4) = \frac{4}{6\sqrt{4^2 + 6^2}} - \frac{1}{10} \approx -0.0075 \dots$$

$$t'(5) = \frac{5}{6\sqrt{5^2 + 6^2}} - \frac{1}{10} \approx 0.0066 \dots$$

x	4	$\frac{9}{2}$	5
t'	-	0	+
t			

iv. (1 mark)

$$\begin{aligned} x &= \frac{9}{2} \\ t &= \frac{\sqrt{36 + 4.5^2}}{6} + \frac{10 - 4.5}{10} \\ &= \frac{9}{5} = 1.8 \text{ h} \end{aligned}$$

Hence he reaches the store in 1 hr 48 min.